HexOct knitting: the Fibonacci swatch

Stacking ABACBCACBABCA, 24 stitches by 85 rows



DICTO · knitting swatch of 31 October 2026

The idea is DICTO's: the knitted fabric is an exact edge-on slice through a stacking of balls, coloured by the pieces between the balls. Chart computed exactly with Claude (Anthropic), generator hexknit.

How to knit every swatch

- Stockinette. Stitches 1 and the last are edge stitches, worked in the chart colour, not slipped, so the edges show the bias steps.
- Colour changes: intarsia with small bobbins, twisting at every change; carry a colour along the back only where its next block is 1–2 stitches away.
- Every bias row keeps the stitch count: one decrease and one increase, each one stitch in from its edge. In the colour runs the increased and decreased stitches are counted as they finish; work each in the colour of its run.
- Bobble (MB), in the colour of the stitch. RS rows: (k1, yo, k1) into the stitch, turn, p3, turn, k3, turn, p3, turn, sl1-k2tog-psso. WS rows: (p1, yo, p1) into the stitch, turn, k3, turn, p3, turn, k3, turn, p3tog, and push it to the RS; if it sits badly, purl the stitch plain and sew a bobble or French knot there afterwards.
- Abbreviations: DG dark gold, LG light gold, DB dark blue, LB light blue. Runs are given in working order (RS from the last stitch to stitch 1, WS from stitch 1 to the last).

Fibonacci: ABACBCACBABCA (chcchchcchcch), 24 stitches × 85 rows

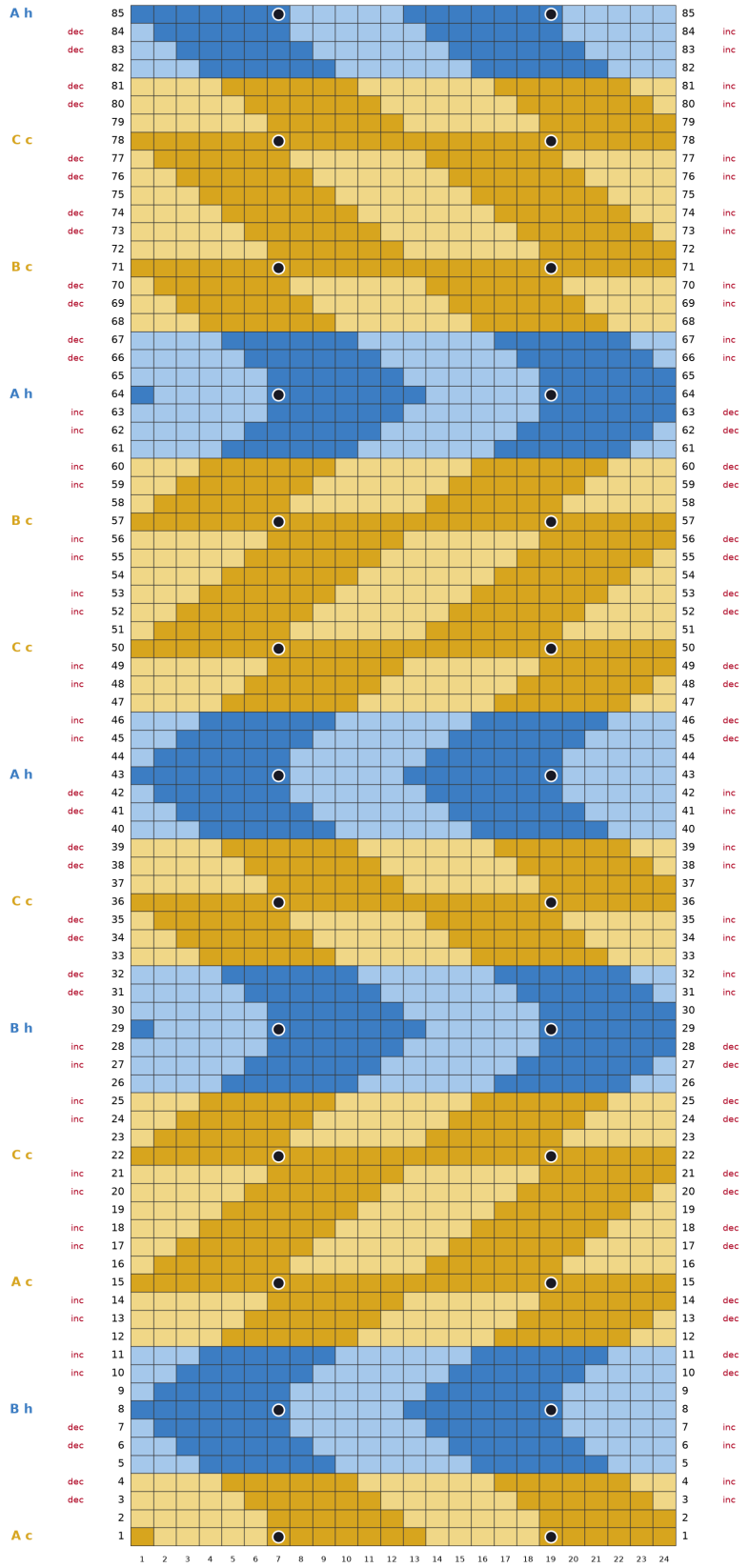
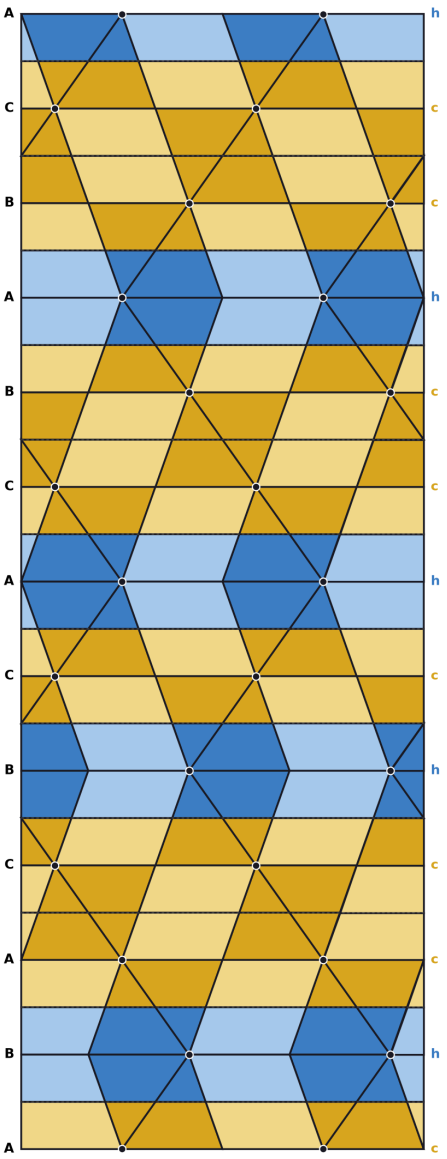


Chart. Seen from the RS, row 1 at the bottom; RS rows read right to left, WS rows left to right. Dots = bobbles. *dec*/*inc*: which edge of a bias row gets the decrease and which the increase.



The exact slice for this window (dots = ball centres; letters = layers; c gold, h blue; dark = tetrahedral, light = octahedral pieces).

DG dark gold: 722 sts · LG light gold: 550 sts · DB dark blue: 394 sts · LB light blue: 374 sts. Total 2040 stitches.

Cast on 24 sts in DG; row 1 is a RS row. **Bias right** (decrease at the left edge, increase at the right, seen from the RS): rows 3, 4, 6, 7, 31, 32, 34, 35, 38, 39, 41, 42, 66, 67, 69, 70, 73, 74, 76, 77, 80, 81, 83, 84. **Bias left**: rows 10, 11, 13, 14, 17, 18, 20, 21, 24, 25, 27, 28, 45, 46, 48, 49, 52, 53, 55, 56, 59, 60, 62, 63. Net shift at the

top: +0 stitches. **Bobbles** on RS rows 1, 15, 29, 43, 57, 71, 85; on WS rows 8, 22, 36, 50, 64, 78. Bind off in LB on the WS after row 85.

Row	Side	Colour runs (working order)	Bias (edge stitches)	Bobble
1	RS	6 DG, 5 LG, 7 DG, 5 LG, 1 DG		MB at st 7, 19
2	WS	6 LG, 6 DG, 6 LG, 6 DG		
3	RS	1 LG, 6 DG, 6 LG, 6 DG, 5 LG	**bias right**: k1, M1, … k2tog, k1	
4	WS	4 LG, 6 DG, 6 LG, 6 DG, 2 LG	**bias right**: p1, p2tog, … M1P, p1	
5	RS	3 LB, 6 DB, 6 LB, 6 DB, 3 LB		
6	WS	2 LB, 6 DB, 6 LB, 6 DB, 4 LB	**bias right**: p1, p2tog, … M1P, p1	
7	RS	5 LB, 6 DB, 6 LB, 6 DB, 1 LB	**bias right**: k1, M1, … k2tog, k1	
8	WS	7 DB, 5 LB, 7 DB, 5 LB		MB at st 7, 19
9	RS	5 LB, 6 DB, 6 LB, 6 DB, 1 LB		
10	WS	2 LB, 6 DB, 6 LB, 6 DB, 4 LB	**bias left**: p1, M1P, … ssp, p1	
11	RS	3 LB, 6 DB, 6 LB, 6 DB, 3 LB	**bias left**: k1, ssk, … M1, k1	
12	WS	4 LG, 6 DG, 6 LG, 6 DG, 2 LG		
13	RS	1 LG, 6 DG, 6 LG, 6 DG, 5 LG	**bias left**: k1, ssk, … M1, k1	
14	WS	6 LG, 6 DG, 6 LG, 6 DG	**bias left**: p1, M1P, … ssp, p1	
15	RS	24 DG		MB at st 7, 19
16	WS	1 LG, 6 DG, 6 LG, 6 DG, 5 LG		
17	RS	4 LG, 6 DG, 6 LG, 6 DG, 2 LG	**bias left**: k1, ssk, … M1, k1	
18	WS	3 LG, 6 DG, 6 LG, 6 DG, 3 LG	**bias left**: p1, M1P, … ssp, p1	
19	RS	2 LG, 6 DG, 6 LG, 6 DG, 4 LG		
20	WS	5 LG, 6 DG, 6 LG, 6 DG, 1 LG	**bias left**: p1, M1P, … ssp, p1	
21	RS	6 DG, 6 LG, 6 DG, 6 LG	**bias left**: k1, ssk, … M1, k1	
22	WS	24 DG		MB at st 7, 19
23	RS	5 LG, 6 DG, 6 LG, 6 DG, 1 LG		
24	WS	2 LG, 6 DG, 6 LG, 6 DG, 4 LG	**bias left**: p1, M1P, … ssp, p1	
25	RS	3 LG, 6 DG, 6 LG, 6 DG, 3 LG	**bias left**: k1, ssk, … M1, k1	
26	WS	4 LB, 6 DB, 6 LB, 6 DB, 2 LB		
27	RS	1 LB, 6 DB, 6 LB, 6 DB, 5 LB	**bias left**: k1, ssk, … M1, k1	
28	WS	6 LB, 6 DB, 6 LB, 6 DB	**bias left**: p1, M1P, … ssp, p1	
29	RS	6 DB, 5 LB, 7 DB, 5 LB, 1 DB		MB at st 7, 19
30	WS	6 LB, 6 DB, 6 LB, 6 DB		
31	RS	1 LB, 6 DB, 6 LB, 6 DB, 5 LB	**bias right**: k1, M1, … k2tog, k1	
32	WS	4 LB, 6 DB, 6 LB, 6 DB, 2 LB	**bias right**: p1, p2tog, … M1P, p1	
33	RS	3 LG, 6 DG, 6 LG, 6 DG, 3 LG		
34	WS	2 LG, 6 DG, 6 LG, 6 DG, 4 LG	**bias right**: p1, p2tog, … M1P, p1	
35	RS	5 LG, 6 DG, 6 LG, 6 DG, 1 LG	**bias right**: k1, M1, … k2tog, k1	
36	WS	24 DG		MB at st 7, 19
37	RS	6 DG, 6 LG, 6 DG, 6 LG		
38	WS	5 LG, 6 DG, 6 LG, 6 DG, 1 LG	**bias right**: p1, p2tog, … M1P, p1	
39	RS	2 LG, 6 DG, 6 LG, 6 DG, 4 LG	**bias right**: k1, M1, … k2tog, k1	
40	WS	3 LB, 6 DB, 6 LB, 6 DB, 3 LB		
41	RS	4 LB, 6 DB, 6 LB, 6 DB, 2 LB	**bias right**: k1, M1, … k2tog, k1	
42	WS	1 LB, 6 DB, 6 LB, 6 DB, 5 LB	**bias right**: p1, p2tog, … M1P, p1	
43	RS	5 LB, 7 DB, 5 LB, 7 DB		MB at st 7, 19
44	WS	1 LB, 6 DB, 6 LB, 6 DB, 5 LB		
45	RS	4 LB, 6 DB, 6 LB, 6 DB, 2 LB	**bias left**: k1, ssk, … M1, k1	
46	WS	3 LB, 6 DB, 6 LB, 6 DB, 3 LB	**bias left**: p1, M1P, … ssp, p1	
47	RS	2 LG, 6 DG, 6 LG, 6 DG, 4 LG		
48	WS	5 LG, 6 DG, 6 LG, 6 DG, 1 LG	**bias left**: p1, M1P, … ssp, p1	
49	RS	6 DG, 6 LG, 6 DG, 6 LG	**bias left**: k1, ssk, … M1, k1	
50	WS	24 DG		MB at st 7, 19
51	RS	5 LG, 6 DG, 6 LG, 6 DG, 1 LG		
52	WS	2 LG, 6 DG, 6 LG, 6 DG, 4 LG	**bias left**: p1, M1P, … ssp, p1	

53	RS	3 LG, 6 DG, 6 LG, 6 DG, 3 LG	**bias left**: k1, ssk, … M1, k1	
54	WS	4 LG, 6 DG, 6 LG, 6 DG, 2 LG		
55	RS	1 LG, 6 DG, 6 LG, 6 DG, 5 LG	**bias left**: k1, ssk, … M1, k1	
56	WS	6 LG, 6 DG, 6 LG, 6 DG	**bias left**: p1, M1P, … ssp, p1	
57	RS	24 DG		MB at st 7, 19
58	WS	1 LG, 6 DG, 6 LG, 6 DG, 5 LG		
59	RS	4 LG, 6 DG, 6 LG, 6 DG, 2 LG	**bias left**: k1, ssk, … M1, k1	
60	WS	3 LG, 6 DG, 6 LG, 6 DG, 3 LG	**bias left**: p1, M1P, … ssp, p1	
61	RS	2 LB, 6 DB, 6 LB, 6 DB, 4 LB		
62	WS	5 LB, 6 DB, 6 LB, 6 DB, 1 LB	**bias left**: p1, M1P, … ssp, p1	
63	RS	6 DB, 6 LB, 6 DB, 6 LB	**bias left**: k1, ssk, … M1, k1	
64	WS	1 DB, 5 LB, 7 DB, 5 LB, 6 DB		MB at st 7, 19
65	RS	6 DB, 6 LB, 6 DB, 6 LB		
66	WS	5 LB, 6 DB, 6 LB, 6 DB, 1 LB	**bias right**: p1, p2tog, … M1P, p1	
67	RS	2 LB, 6 DB, 6 LB, 6 DB, 4 LB	**bias right**: k1, M1, … k2tog, k1	
68	WS	3 LG, 6 DG, 6 LG, 6 DG, 3 LG		
69	RS	4 LG, 6 DG, 6 LG, 6 DG, 2 LG	**bias right**: k1, M1, … k2tog, k1	
70	WS	1 LG, 6 DG, 6 LG, 6 DG, 5 LG	**bias right**: p1, p2tog, … M1P, p1	
71	RS	24 DG		MB at st 7, 19
72	WS	6 LG, 6 DG, 6 LG, 6 DG		
73	RS	1 LG, 6 DG, 6 LG, 6 DG, 5 LG	**bias right**: k1, M1, … k2tog, k1	
74	WS	4 LG, 6 DG, 6 LG, 6 DG, 2 LG	**bias right**: p1, p2tog, … M1P, p1	
75	RS	3 LG, 6 DG, 6 LG, 6 DG, 3 LG		
76	WS	2 LG, 6 DG, 6 LG, 6 DG, 4 LG	**bias right**: p1, p2tog, … M1P, p1	
77	RS	5 LG, 6 DG, 6 LG, 6 DG, 1 LG	**bias right**: k1, M1, … k2tog, k1	
78	WS	24 DG		MB at st 7, 19
79	RS	6 DG, 6 LG, 6 DG, 6 LG		
80	WS	5 LG, 6 DG, 6 LG, 6 DG, 1 LG	**bias right**: p1, p2tog, … M1P, p1	
81	RS	2 LG, 6 DG, 6 LG, 6 DG, 4 LG	**bias right**: k1, M1, … k2tog, k1	
82	WS	3 LB, 6 DB, 6 LB, 6 DB, 3 LB		
83	RS	4 LB, 6 DB, 6 LB, 6 DB, 2 LB	**bias right**: k1, M1, … k2tog, k1	
84	WS	1 LB, 6 DB, 6 LB, 6 DB, 5 LB	**bias right**: p1, p2tog, … M1P, p1	
85	RS	5 LB, 7 DB, 5 LB, 7 DB		MB at st 7, 19